

Inspection of the *MV Example* to determine the level of biofouling and detect Invasive Marine Species

Date: 19 September, 2018

Location: Singapore Cruise Centre, Singapore

Prepared for: Company X



Report prepared by

Company X

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EXECUTIVE SUMMARY

Company X is managing the 221 m cruise ship *MV Example* on behalf of X Cruises. In the 2018-19 cruise season the vessel will operate in the waters of Australia, Southeast Asia, Asia and New Zealand. Given the *MV Example* will visit several jurisdictions in Australia and New Zealand where management of biofouling or Invasive Marine Species (IMS) is a regulatory requirement, Company X engaged the services of X to conduct an in-water biofouling inspection of the external hull, and to assist in managing any observed biofouling to reduce the risk of transferring IMS of concern to destination ports. The biofouling inspection was conducted while the *MV Example* was alongside at the Singapore Cruise Centre, Singapore on 19 September, 2018.

During the in-water biofouling inspection of the *MV Example*, the hull was found to be in good condition with biofouling on the general hull typically consisting of primary biofouling composed of diatomaceous slime and a waterline band of green algae. Secondary biofouling on the hull plating was largely restricted to the forward sections of the waterline (forward of Frame 198), the aft sections of waterline (aft of Frame 14) and the starboard quarter of the transom. This secondary biofouling was associated with a band of green algae extending to approximately 5 m depth and consisted of scattered barnacles (c.f. *Amphibalanus reticulatus* <5%) and tubeworms (*Hydroides* sp. <1%), with biofouling concentrated on weld seams. Some scattered secondary biofouling consisting of scattered individual acorn barnacles and tubeworms (<1%) was detected over other areas of the hull. A thorough inspection of niche areas determined that biofouling in niche areas was typically restricted to secondary biofouling composed of acorn barnacles (c.f. *Amphibalanus reticulatus*), tubeworms (*Hydroides* sp.), hydroids and bryozoans. In most cases biofouling in niche areas represented less than 5% coverage. However, biofouling associated with overboard pipes, draft markers (forward and aft), thruster propeller blades, anodes associated with the port stabiliser fin, rope guard voids and rudder stock areas exceeded this level. Secondary biofouling in some niche areas represented up to 90% coverage. Tertiary biofouling composed of scattered individuals of mussels (*Mytilus* sp.) was restricted to the rudder stock areas and a single individual associated with a port rope cutter blade (individual removed). The Inspector achieved a high level of confidence that IMS of concern would have been detected if present and that biofouling in excess of the Craft Risk Management Standard (CRMS) thresholds were accurately described by the inspection.

No macroscopic IMS of concern listed in regulatory requirements pertaining to Australian state legislation were detected in association with the *MV Example*. Thus, at the conclusion of the inspection, the vessel was determined to be compliant with the regulatory obligations of the Northern Territory, Queensland, New South Wales, Victoria and Tasmania. However, it should be recognised that microscopic life-history stages may have been present and not detected, and that individuals of IMS of concern may settle on the vessel's submerged hull subsequent to the inspection. To minimise the chance of infection the *MV Example* should limit exposure to IMS propagules by minimizing time spent in coastal waters.

An assessment of the *MV Example* determined that the vessel is not presently compliant with the New Zealand CRMS “long-stay” threshold. Furthermore, various hull and niche locations have biofouling levels that are considered non-compliant with the CRMS “short-stay” thresholds. If Company X intends to demonstrate that the *MV Example* is compliant with the CRMS “short-stay” thresholds, some level of in-water cleaning is necessary, followed by an in-water biofouling inspection to determine that the vessel is compliant with appropriate biofouling thresholds prior to entering New Zealand territorial waters.

X has been advised that Company X is considering undertaking an in-water clean to remove problematic biofouling in Hong Kong (3-5 November, 2018), Laem Chabang (17-19 November, 2018) or Singapore (23 November, 2018). While it may be possible to facilitate an opportunistic clean in Singapore on 23 November, 2018, the risks associated with the short turn around and the possibility of simultaneous operations hampering the dive effort (e.g. bunkering, passenger operations etc.) may result in a high risk of the scope of work not being completed. Unfortunately, X has limited experience with commercial dive operators in Laem Chabang or Hong Kong but it is likely that divers based in these locations will not be trained in biofouling management to a similar standard to those in Singapore. An alternative may be to complete the cleaning in either Laem Chabang or Hong Kong and then engage a commercial dive company in Singapore to conduct a final inspection on 23 November with divers standing by to clean any residual biofouling not removed during the previous dive operation.

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1. INTRODUCTION

1.1 Invasive Marine Species Background

Invasive Marine Species (IMS) are widely recognised as one of the most significant threats to marine ecosystems worldwide, along with overfishing and human-driven atmospheric and oceanic alterations. Shallow coastal marine environments in particular, are thought to be amongst the most heavily invaded ecosystems on earth, which largely reflects the accidental transport of IMS by international shipping. A number of human-mediated vectors can transport IMS outside their native ranges. For instance ocean-going vessels can transport marine species in ballast water, as biofouling attached to submerged immersible equipment, within internal seawater systems and/or on the exterior of the hull. For the past three decades, ballast water was considered the primary vector responsible for the dispersal of IMS around the world (Carlton 1985; Thresher et al. 1999; Eldredge and Carlton 2002). However, recent research suggests the role of ballast water was probably overstated, and up to 69% of these introductions may have occurred via biofouling (Hewitt et al. 1999, 2004; Hewitt and Campbell 2010). To prevent further introductions, and limit the spread of IMS, shipping requires management for both ballast water and biofouling.

Around the globe there are approximately 2,000 non-indigenous species that are known to have been introduced to marine or estuarine systems. Some of which have had catastrophic ecological, economic, human health and social/cultural consequences (Carlton 1996, 2001; Pimental et al. 2000; Hewitt et al. 2011). In Australian marine waters there are approximately 450 established marine species that are either non-indigenous, or whose origins are inconclusive (Hewitt et al. 2011). IMS that survive and establish reproductive populations in regions where they did not previously occur, often have no known impacts, although some species can cause major damage and are referred to as 'marine pests' or 'IMS of concern'. Some IMS can facilitate the introduction of additional species and their cumulative impacts can significantly undermine the structure, function and persistence of native ecosystems. It is predicted that each year approximately three to four additional IMS will continue to establish in Australian waters unless marine biosecurity management improves.

Once established, IMS are often difficult or impossible to eradicate. Most attempts to eradicate IMS are logistically challenging, complex, costly and unsuccessful. The pragmatic approach to protecting marine environments from the impacts of IMS, is to prevent their introduction and spread by human-mediated transport. This approach to marine biosecurity provides the fundamental rationale for most global and regional management initiatives.

2. BIOFOULING AND IMS MANAGEMENT REGULATIONS RELEVANT TO THE *MV Example*

2.1 The *MV Example*

Company X is managing the 221 m cruise ship *MV Example* on behalf of Viking Ocean Cruises. In the 2018-19 cruise season the vessel will operate in the waters of Australia, Southeast Asia, Asia and New Zealand. Given the *MV Example* will visit several jurisdictions where management of biofouling or Invasive Marine Species (IMS) is a regulatory requirement, Company X has engaged the services of X to conduct an in-water biofouling inspection of the external hull, and to assist in managing any observed biofouling to reduce the risk of transferring IMS of concern to destination ports. Furthermore, Company X requested that X offer advice on what diver-cleaning measures can be adopted during subsequent port stays to remove any excessive biofouling.

The biofouling inspection was conducted while the *MV Example* was alongside at the Singapore Cruise Centre, Singapore on 19 September, 2018. This report documents the outcome of the inspection and provides advice to Company X on the best approach to manage the vessel to demonstrate compliance with regulatory obligations.

2.2 Relevant Biofouling Regulations

Subsequent to the IMS inspection the *MV Example* is scheduled to visit a number of destination ports in Asia, Australia and New Zealand. A summary of destination ports between September 2018 and March 2019 is provided in **Figure 1** with ports marked by a star indicating locations with mandatory regulations in place providing authorities with powers in relation to either the presence of specific IMS (i.e. Australia) or biofouling beyond a specific threshold (i.e. New Zealand). Given the intended itinerary of the *MV Example*, three key biofouling management obligations need to be considered:

- 1) Mandatory compliance with state legislation in Australia
- 2) Mandatory compliance with the New Zealand *Biosecurity Act 1993* and the Craft Risk Management Standard for biofouling.



Figure 1 A summary of key destination ports for the *MV Example* in the 2018-2019 cruise seasons. Destinations marked with a ★ indicate ports subject to Australian state regulatory obligations governing the import of listed “noxious” aquatic species (see Section 2.3). Destinations marked with a ★ indicate ports subject to the Craft Risk Management Standard governing acceptable thresholds for biofouling (see Section 2.4).

2.3 Australian Marine Biosecurity Management

A key role of the Commonwealth's Department of Agriculture and Water Resources (DAWR) is to safeguard Australia's favourable human, animal and plant health status. Previously, this was regulated under the *Quarantine Act 1908* and its subordinate legislation (i.e. *Quarantine Regulations 2000* and the *Quarantine Proclamation 1998*). However, these have now been superseded by the *Biosecurity Act 2015* which was introduced on 16 June, 2016. The key legislative change between the *Quarantine Act 1908* and the *Biosecurity Act 2015* is the jurisdictional shift of the DAWR from 200 nm to 12 nm (e.g. Australian territory).

At this stage there are no intentions within the *Biosecurity Act 2015* to manage biofouling on facilities or vessels entering Australian territory. However, despite there being no mandatory Commonwealth biofouling regulations in place, the Commonwealth previously released guidance documents for national biofouling management as part of the 'national system for the prevention and management of marine pest incursions' covering various vessel classes and industry sectors. These notably include guidelines for the petroleum production and exploration industry. For further details see (<http://marinepests.gov.au/petroleum/managing-biofouling>).

Considering the lack of specific biofouling management regulations at the Australian Commonwealth level, states and Territories within Australia have developed a range of regulatory tools to protect their state waters from the threats associated with IMS. While all States and Territories of Australia have some powers to act against the introduction of IMS of concern, some States/Territories are more active than others, and in some instances legislation that implies specific powers to state authorities are not actively enforced. A review of biofouling species specifically prohibited in state water that will be visited by the *MV Example* are summarised in **Table 1** along with a reference to the relevant regulatory mechanisms. In all cases these powers are confined to state waters (i.e. within 3 nm seaward of the Territorial Sea baseline).

It should be noted that no specific regulations regarding vessel inspections are in place in any of the states to be visited by the *MV Example*. Thus while state authorities have the power to act in relation to a verified biosecurity risk, there is presently no formalised reporting process or compliance checks implemented.



Table 1 “Noxious” or “Prohibited” biofouling species listed under relevant Australian state legislation. NT Species: Noxious fish refer to those listed in Schedule 1 of the *Fisheries Regulations 2012*; QLD Species: Prohibited matter species are listed in Schedule 1 of the *Biosecurity Act 2014*; NSW Species: Prohibited matter listed in Schedule 2 of the *Biosecurity Act 2015*. VIC Species: Noxious Aquatic Species declared under Section 75 of the *Fisheries Act 1995* – updated list provided online: <https://vfa.vic.gov.au/operational-policy/pests-and-diseases/noxious-aquatic-species-in-victoria> (last updated 2010); TAS species: Noxious Fish listed in Part 4, Section 20 of *Fisheries (General and Fees) Regulations 2006*.

Phylum	Scientific Name	Common Name	NT	QLD	NSW	VIC	TAS
Algae	<i>Caulerpa taxifolia</i> (exotic strains only)	Green macroalga	x	x		x	x
Algae	<i>Codium fragile fragile</i>	Green macroalga	x	x			
Algae	<i>Grateloupia turuturu</i>	Red macroalga	x	x	x		
Algae	<i>Sargassum muticum</i>	Asian seaweed	x	x	x		
Algae	<i>Undaria pinnatifida</i>	Japanese seaweed	x	x		x	x
Annelida	<i>Alitta succinea</i>	Pile worm	x				
Annelida	<i>Boccardia proboscidea</i>	Spionid polychaete	x				
Annelida	<i>Euchone limnicola</i>	Sabellid polychaete worm	x				
Annelida	<i>Marenzelleria</i> spp.	Red gilled mudworm	x	x			
Annelida	<i>Polydora websteri</i>	Mudworm	x				
Annelida	<i>Polydora cornuta</i>	Spionid polychaete	x				
Annelida	<i>Sabella spallanzanii</i>	European fan worm	x	x		x	x
Ascideacea	<i>Ciona intestinalis</i>	Sea vase	x				
Ascideacea	<i>Didemnum</i> spp. (exotic invasive strains only)	Colonial sea squirt	x	x			
Ascideacea	<i>Didemnum vexillum</i>	Colonial sea squirt	x	x	x		
Ascideacea	<i>Styela clava</i>	Clubbed tunicate	x				

Phylum	Scientific Name	Common Name	NT	QLD	NSW	VIC	TAS
Cnidaria	<i>Mnemiopsis leidyi</i>	Comb jelly	x	x	x		
Crustacea	<i>Balanus improvisus</i>	Barnacle	x	x	x		
Crustacea	<i>Eriocheir</i> spp.	Chinese mitten crab	x	x	x		
Crustacea	<i>Hemigrapsus sanguineus</i>	Japanese/Asian shore crab	x	x	x		
Crustacea	<i>Petrolisthes elongatus</i>	New Zealand half shell crab	x				
Crustacea	<i>Hemigrapsus takanoi/penicillatus</i>	Pacific crab	x	x	x		
Crustacea	<i>Charybdis japonica</i>	Lady crab	x	x	x		
Crustacea	<i>Carcinus maenas</i>	European green crab	x	x			x
Crustacea	<i>Procambarus clarkii</i>	Red swamp crayfish			x	x	
Echinoderm	<i>Asterias amurensis</i>	Northern Pacific seastar	x	x	x	x	x
Teleost	<i>Neogobius melanostomus</i>	Round goby	x	x	x		
Teleost	<i>Siganus rivulatus</i>	Marbled spinefoot, rabbit fish	x	x	x		
Holoplankton	<i>Alexandrium monilatum</i>	Toxic dinoflagellate	x	x			
Holoplankton	<i>Dinophysis norvegica</i>	Toxic dinoflagellate	x	x			
Holoplankton	<i>Pfiesteria piscicida</i>	Toxic dinoflagellate	x	x			
Holoplankton	<i>Chaetoceros concavicornis</i>	Centric diatom	x	x			
Holoplankton	<i>Chaetoceros convolutus</i>	Centric diatom	x	x			
Holoplankton	<i>Pseudo-nitzschia seriata</i>	Pennate diatom	x	x			
Mollusca	<i>Musculista senhousia</i>	Asian bag mussel	x	x	x		
Mollusca	<i>Mytilopsis sallei</i>	Black striped mussel	x	x	x	x	x
Mollusca	<i>Perna perna</i>	Brown mussel	x	x	x		

Phylum	Scientific Name	Common Name	NT	QLD	NSW	VIC	TAS
Mollusca	<i>Perna viridis</i>	Asian green mussel	x	x	x		
Mollusca	<i>Perna canaliculus</i>	New Zealand green mussel			x		
Mollusca	<i>Corbula (Potamocorbula) amurensis</i>	Asian clam, brackish-water corbula	x	x	x		
Mollusca	<i>Ensis directus</i>	Jack-knife clam	x	x	x		
Mollusca	<i>Mya arenaria</i>	Soft shell clam	x	x	x		
Mollusca	<i>Theora lubrica</i>	Asian semelid bivalve	x				
Mollusca	<i>Varicorbula gibba</i>	European clam	x	x	x		
Mollusca	<i>Crepidula fornicata</i>	American slipper limpet	x	x	x		
Mollusca	<i>Rapana venosa</i> (syn <i>Rapana thomasi</i>)	Rapa whelk	x	x	x		
Mollusca	<i>Magalana gigas</i>	Pacific Oyster		x			
Mollusca	<i>Maoricolpus roseus</i>	New Zealand screw shell	x	x			

2.4 New Zealand Biofouling Management and the Craft Risk Management Standard

The biosecurity threat to New Zealand coastal waters posed by IMS entering territorial waters in association with biofouling is presently regulated via *Biosecurity Act 1993*. Specifically, Section 33(1)(b) of the Act states that:

- (1) *Where there are any risk goods on board or attached to the outside of a craft that has entered New Zealand territory from outside New Zealand territory, an inspector may direct the master or other person in charge of the craft to ...*
- (b) *move the craft outside New Zealand territory (immediately, or within a period specified by the inspector);*

As such, the identification of a vessel with biofouling containing any non-indigenous marine species considered as posing a risk to the New Zealand's marine resources, economy, environment, and/or people's health the New Zealand Ministry for Primary Industries (NZ MPI) has the power to deny the vessel entry into New Zealand's territorial waters or turn the vessel around and order its immediate departure.

Though the *Biosecurity Act 1993* provides broad powers for New Zealand to act upon identified biofouling related threats, the NZ MPI developed their CRMS to better protect the waters of New Zealand from the threat posed by non-indigenous species arriving via biofouling (see Appendix 1). The CRMS sets clear standards for the management of biofouling risk associated with all vessels (sea craft) that are entering New Zealand territorial waters (within 12 nm from the coastal baseline). These standards represent the most comprehensive management regime for biofouling related introductions applied to any state in the world¹. The CRMS was designed with the intention to compliment the 2011 IMO Biofouling Guidelines. As such, the CRMS adopts an approach of managing the general level of biofouling as opposed to a species-specific approach implemented by other jurisdictions such as those in Western Australia and other regions.

In essence, the CRMS will require that all vessels visiting New Zealand territorial waters have a "clean hull", with the acceptable thresholds for this status being defined according to the intended length of stay in New Zealand territorial waters. Two categories of vessels are distinguished in the CRMS:

- 1) **'Long-stay vessels'** means those vessels intending to remain in New Zealand for 21 days or longer, or those vessels intending to visit areas other than those designated under Section 37 of the *Biosecurity Act 1993* as 'Places of First Arrival'.

¹ Note that some localised high value areas such as Barrow Island in Western Australia, the Hawaiian National Monument and the New Zealand Subantarctic Islands are managed to similar or more stringent standards.

- 2) **‘Short-stay vessels’** means those vessels intending to remain in New Zealand for 20 days or less and to only visit places designated under section 37 of the Act as ‘Places of First Arrival’. These vessels generally remain under ‘biosecurity surveillance’ while in New Zealand territory rather than becoming fully cleared of risk goods.

Thresholds for the definition of a “clean hull” for each of these vessel categories are provided in Table 2.

Table 2. Allowable biofouling threshold relative to intended length of stay in New Zealand waters.

Length of stay	Hull Area	Allowable Biofouling
0 - 20 days (Short -stay)	All hull surfaces	Slime layer; Gooseneck barnacles.
	Wind and water line	Algae growth of unrestricted cover and no more than 50 mm in frond, filament or beard length; Brown and red algal growth of no more than 4 mm in length; Incidental (maximum of 1%) coverage of one organism type of either tubeworms, bryozoans or barnacles, occurring as: • Isolated individuals or small clusters; and • A single species, or what appears to be the same species.
	Hull area	Algal growth occurring as: • No more than 4 mm in length; and • Continuous strips and/or patches of no more than 50 mm in width. Incidental (maximum of 1%) coverage of one organism type of either tubeworms, bryozoans or barnacles, occurring as: • Isolated individuals or small clusters that have no algal overgrowth; and • A single species, or what appears to be the same species.
	Niche areas	Algal growth occurring as: • No more than 4 mm in length; and • Continuous strips and/or patches of no more than 50 mm in width. Scattered (maximum of 5%) coverage of one organism type of either tubeworms, bryozoans or barnacles, occurring as: • Widely spaced individuals and/or infrequent, patchy clusters that have no algal overgrowth; and • A single species, or what appears to be the same species; and

Length of stay	Hull Area	Allowable Biofouling
		Incidental (maximum of 1%) coverage of a second organism type of either tubeworms, bryozoans or barnacles, occurring as: - Isolated individuals or small clusters that have no algal overgrowth; and - A single species, or what appears to be the same species.
>20 days (long-stay)	All hull surfaces	Slime layer; Gooseneck barnacles.

The CRMS sets out three Acceptable Measures that are considered as sufficient for meeting the “clean hull” requirement:

- 1) Cleaning before visit to New Zealand**, (or immediately on arrival in a facility or by a system, approved by NZ MPI). All biofouling must be removed from all parts of the hull and this must be carried out less than 30 days before arrival to New Zealand or within 24 hours after time of arrival.
- 2) Continual Maintenance using best practice** including: application of appropriate antifoul coatings; operation of Marine Growth Prevention Systems (MGPS) within sea-chests; and in-water inspections with biofouling removal as required. Following the 2011 IMO Biofouling Guidelines is recognised as an example of best practice.
- 3) Application of Approved Treatments.** Treatments are approved and listed under the Approved Biosecurity Treatments MPI-STD- ABTRT.

Additional information regarding the three acceptable measures is provided in the draft Guidance Document for the CRMS - Biofouling on Vessels Arriving to New Zealand. Furthermore, as an alternative to the Acceptable Measures above, a vessel operator may submit, for NZ MPI approval, a Craft Risk Management Plan (CRMP) under Section 24K of the *Biosecurity Act 1993* (which includes steps that will be taken to reduce risk to the equivalent degree as meeting the requirements of this standard).

To demonstrate that an operator has effectively implemented one of the Acceptable Measures to meet the “clean hull” requirement, operators will be required to submit to NZ MPI the following information:

- 1) Prior to arriving in New Zealand all operators of vessels must provide to MPI the following information (via the Advanced Notice of Arrival and associated documents).*
 - a. Intended length of stay within New Zealand territory and intentions in respect of places to be visited;*
 - b. Whether the vessel has spent any extended periods mainly stationary in a single location;*

- c. If the vessel is coming in to undergo biofouling cleaning on arrival, any formal arrangement for cleaning or treatment and whether they will be undertaken immediately upon arrival (within 24 hours);*
 - d. What measures have been or will be used to meet the requirements of the standard; or*
 - e. Whether the operator has chosen to operate an NZ MPI approved Craft Risk Management Plan (CRMP) as an alternative to meeting the requirements of the standard.*
- 2) The following information (if relevant) must be held on the vessel and provided to MPI in an appropriate form if requested. (This is in addition to information to be provided prior to arrival).*
 - a. Information on the antifouling regime and any marine growth prevention systems used;*
 - b. Whether applying the IMO Biofouling Guidelines, including employing a biofouling management plan showing the hull maintenance and inspection regime and records of biofouling management kept, preferably consistent with the template in the IMO guidelines;*
 - c. If applicable to the vessel, its latest International Anti-fouling System Certificate or International Anti-fouling System Declaration;*
 - d. Date and reporting from the latest hull biofouling inspection.*

3. BACKGROUND

3.1 Vessel Details

OWNER:	X Company
CONTACT PERSON:	X Client
VESSEL/STRUCTURE'S NAME:	<i>MV Example</i>
FLAG:	Norway
PORT OF REGISTRY:	Bergen
VESSEL/STRUCTURE TYPE:	Cruise Ship
IMO NUMBER:	97XXXXX
CALL SIGN:	XXXXX
MARITIME MOBILE SERVICE IDENTITY (MMSI):	XXXXXXXXX
DISPLACEMENT (TONNES):	47,861
LENGTH (M):	221
BREADTH (M):	34
MAXIMUM DRAFT (M):	6.7
GENERAL ARRANGEMENT:	Provided in Appendix 1.

3.2 Antifouling and Maintenance Details

2.5.1 Antifouling coating

DATE OF APPLICATION:	25 September, 2017
TIME ELAPSED SINCE ² :	360 Days (~12 months)
COATING TYPE:	Jotun SeaQuantum Ultra S
LIFE EXPECTANCY:	60 months
LOCATION OF APPLICATION:	All submerged surfaces
Antifouling Coating certification is included in Appendix 2.	

4. INSPECTION DETAILS AND METHODS

DATES:	19 September, 2018
PRIMARY INSPECTOR:	X
SECONDARY INSPECTOR:	N/A

² "Time elapsed since" is the period of time between the previous dry-dock/antifoul or inspection, and the current inspection.

LOCATION:	Singapore Cruise Centre, Singapore	
INSPECTION TYPE:	☒ In-water inspection	☐ Dry-dock inspection
AIM OF INSPECTION:	☒ Locate IMS of concern ³	☒ Determine level of biofouling ⁴
INSPECTION CLASS:	☐ Inspection within Australia	☐ Inspection and clean within Australia
	☒ International inspection	☐ International inspection and clean
REASON FOR INSPECTION:	☐ Required by contractor. Contractor:	
	☒ Other: Required to assess regulatory compliance with NZ Craft Risk Management Standard and east coast Australian state regulations.	
AREAS INSPECTED:	☐ Topside immersible equipment ☐ Internal seawater systems	☒ External hull ☐ Ballast tanks
INSPECTION EQUIPMENT USED:	☒ Digital camera ☒ X record sheet ☒ Scraper and collection bags	☒ Microscope ☐ Endoscope ☒ Taxonomic guides
SAMPLES PRESERVED:	☒ Yes ☐ No Comments: Samples of barnacles, tubeworms and mussels retained in 70% ethanol.	

³ The detection of IMS of concern such as microscopic algae and juvenile life stages was not possible during the inspection due to their size.

⁴ Primary, secondary and tertiary levels of biofouling refer to a simplistic tool to measure the degree of biofouling present and likelihood of IMS of concern being present (See Appendix 3).

4.1 Supporting documents completed and made available to inspector

Confirmed	Document
☐ Yes ☐ No ☒ N/A	001 Client Enquiry Questionnaire
☐ Yes ☐ No ☒ N/A	001a Vessel Specific Questionnaire
☒ Yes ☐ No ☐ N/A	General Arrangement
☒ Yes ☐ No ☐ N/A	Antifouling Certificate
☐ Yes ☐ No ☒ N/A	Docking Plan
☐ Yes ☐ No ☒ N/A	Internal Seawater System/s schematic diagrams
☒ Yes ☐ No ☐ N/A	Voyage History
☐ Yes ☐ No ☒ N/A	Marine Growth Prevention System Logbook
☒ Yes ☐ No ☐ N/A	011a Diver Quality Inspection and Cleaning Assurance Statement

4.2 External Hull Inspection

START DATE AND TIME:	11:50 hours, 19 September, 2018	
FINISH DATE AND TIME:	17:00 hours, 19 September, 2018	
DIVE COMPANY:	X	
DIVE VESSEL:	X	
DIVE SUPERVISOR:	X	
DIVER:	X (Dive 1), X (Dive 2)	
DIVING AND INSPECTION EQUIPMENT USED:	☒ Surface supply ☒ Digital Camera ☐ Endoscope ☐ HP Blasters ☐ Hand Scrapers	☒ Communications and CCTV ☒ Scraper and catch-bag ☐ Rotational brush unit ☐ Pneumatic Polishing Tools ☐ Other:
ALL HULL LOCATIONS/NICHE AREAS INSPECTED:	☐ Yes	☒ No
HULL LOCATIONS/NICHE AREAS NOT INSPECTED:	Sea chests and tunnel thruster inspected through gratings only. Only a selection of overboard discharge pipes inspected. Stabiliser fins inspected in the retracted position.	
RANGE/AVERAGE WATER CLARITY DURING FINAL INSPECTION:	2 m	
TIDAL CURRENTS DURING FINAL INSPECTION:	<1 knots	
INSPECTION RECORDED ON DVD:	☒ Yes	☐ No
COMMENTS: The biofouling inspection was initially delayed while a bunkering vessel was mooring alongside the <i>MV Example</i> . Following the commencement of dive operations at 11:50 hours all effort was made to conduct a thorough inspection with available time. The biofouling inspector obtained a high level of confidence that the vessel was accurately surveyed during the inspection.		

5. RESULTS

An overview of inspection findings relevant to biofouling regulations in Australia and New Zealand are presented in Sections 5.1 and 5.2. Detailed results from the final inspection presented in Sections 5.3. Final inspection photographs are supplied in Appendix 4 as representative images of niche and inspection strata. Images of all areas inspected can be made available on DVD upon request.

5.1 Summary of Biofouling Inspection Results relevant to Australian regulations

Inspection standards	Suspected IMS of Concern Found	Comments
IMS of concern (Australian state regulations)	☐ Yes ☒ No	No macroscopic IMS of concern listed in any Australian state regulations were detected on the <i>MV Example</i> .

5.2 Summary of Biofouling Inspection Results relevant to New Zealand regulations

Inspection standards	Biofouling detected in excess of threshold?	Comments
CRMS Long-stay threshold	☒ Yes ☐ No	Biofouling in excess of the CRMS long-stay threshold included the widespread presence of secondary biofouling (acorn barnacles, tubeworms and bryozoans), and the presence of isolated areas of tertiary biofouling (mussels).
CRMS Short stay threshold	☒ Yes ☐ No	Secondary biofouling outside of the CRMS short-stay thresholds was detected around the vessel waterline, overboard discharge pipes, thruster tunnels (props only), Port stabiliser fin void (anode only), rudders, propellers and rope-guards.

5.3 Final External Hull Inspection: Detailed Results

See Appendix 4 for representative photographs.

Niche Type	No, inspected/ No. present	Biofouling Level (C/P/S/T) ⁵	Biofouling in excess of CRMS detected? (Y/N)	Comments
Bulbous bow	1/1	S	Y	Band of biofouling consisting of green algae and > 5% scattered acorn barnacles and tubeworms extending to 5 m depth.
Waterline	1/1	S	Y	Forward section of waterline (forward of Frame 198), and aft section of waterline (aft of Frame 14) with <5% scattered barnacles and tubeworms in a band extending to approximately 5 m. Secondary biofouling concentrated on weld seams. Transom waterline with <5% scattered barnacles and tubeworms restricted to starboard quarter.
General hull	1/1	S	N	Mostly clean with light biofouling consisting of green algae, diatomaceous slime and scattered acorn barnacles at a density of ~1%.
Bilge keels	2/2	S	N	Mostly clean with diatomaceous slime only. Some small areas of damaged paint (<1%).
Forward Draft markers	2/2	S	Y	Biofouling consisting of <5% acorn barnacles and tubeworms associate with forward draft markers extending to depth of 5 m. Draft markers deeper than 5 m with isolated acorn barnacles (<1%).
Amidships Draft Markers	1/2	S	N	Port side draft markers inspected and found to have biofouling consisting of algae, barnacles (~5%) and tubeworms (~1%). As a niche area this is considered compliant with the CRMS thresholds for short-stay vessels.
Aft Draft Markers	2/2	S	Y	Biofouling consisting of 5 - 10% acorn barnacles and tubeworms associate with forward draft markers extending to depth of 5 m. Draft markers deeper than 5 m with isolated acorn barnacles (<1%).
Overboard pipes	13/?	S	Y	All inspected overboard pipes found to have significant biofouling consisting of 60 – 90% acorn barnacles, tubeworms, bryozoans and hydroids.

⁵ Level of Biofouling: C = Clear, P = Primary, S = Secondary, T = Tertiary. For further details see Appendix 3.

Niche Type	No, inspected/ No. present	Biofouling Level (C/P/S/T) ⁵	Biofouling in excess of CRMS detected? (Y/N)	Comments
Sea chest grates	3/6	S	N	Only non-operational sea chests could be inspected. Biofouling restricted to 1-5% acorn barnacles and tubeworms (<1%), generally associated with bolts or areas of damaged paint. As a niche area this is considered compliant with the CRMS thresholds for short-stay vessels.
Sea chest boxes	3/6	P/S	N	Only non-operational sea chests could be inspected and inspection conducted through grates (gratings not removed). Inside of sea chests appeared clean with biofouling mostly restricted to diatomaceous slime. Port aft low sea chest had isolated acorn barnacles inside (<1%). As a niche area this is considered compliant with the CRMS thresholds for short-stay vessels.
Sounder equipment	3/3	S	N	Mostly clean with some secondary biofouling mostly associated with bolts and consisting of acorn barnacles (<5%) and tubeworms (<1%). As a niche area this is considered compliant with the CRMS thresholds for short-stay vessels.
Bow thrusters	2/2	S	Y	Inspected through gratings only. Gratings with <1% acorn barnacles and tubeworms mostly associated with damages paint surfaces. Thruster tunnels, gear housings and anodes were clean with biofouling restricted to diatomaceous slime. Thruster propellers with ~50% biofouling consisting of acorn barnacles, tubeworms, hydroids and algae.
Stern thruster	1/1	S	Y	Inspected through gratings only. Gratings with <1% acorn barnacles and tubeworms mostly associated with damages paint surfaces. Thruster tunnels and anodes were clean with biofouling restricted to diatomaceous slime. Gear housing with scattered acorn barnacles and tubeworms (<5%). Thruster propellers with ~60-70% biofouling consisting of acorn barnacles, tubeworms, hydroids and algae.

Niche Type	No, inspected/ No. present	Biofouling Level (C/P/S/T) ⁵	Biofouling in excess of CRMS detected? (Y/N)	Comments
Stabilising fins	2/2	S	Y	Inspected in retracted position. Void space and fin/hinge surfaces mostly clean with biofouling restricted to ~5% acorn barnacles and ~1% tubeworms. One anode associated with port side stabiliser fin void with ~50% biofouling consisting of tubeworms, hydroids and acorn barnacles.
Rudders	2/2	S/T	Y	Plating surfaces of rudders clean with no biofouling. Rudder hinges and void spaces with ~60% secondary biofouling consisting of acorn barnacles and tubeworms. Anodes with 20-40% secondary biofouling consisting of acorn barnacles and tubeworms. Non-toxic surfaces associated with top of rudder and rudder stock with 20-60% secondary biofouling consisting of acorn barnacles and tubeworms. Scattered individual mussels (<i>Mytilus</i> sp.) present around rudder stock of both rudders (estimated 10%).
Propellers	2/2	S	N	Blades mostly clean with no biofouling. Scattered dead acorn barnacles associated with base of blades (<5%).
Interface surfaces of 'Promes' propellers	2/2	S	Y	Trailing face of propeller boss with 70-90% secondary biofouling composed primarily of tubeworms with scattered acorn barnacles. Rudder facings with ~10% tubeworms with scattered acorn barnacles.
Propeller shaft and brackets	2/2	C	N	Shaft and bracket surfaces clean with no biofouling.
Rope guards and rope cutter voids (propeller and shaft)	10/10	S/T	Y	Cutter blades and assemblies with <5% secondary biofouling consisting primarily of tubeworms with scattered (<1%) acorn barnacles. A single juvenile mussel (<i>Mytilus</i> sp.) was detected and removed from a port side rope cutter blade associated with the shaft. Void spaces of shafts and propeller rope guards with 40-60% secondary biofouling consisting primarily of tubeworms with scattered (~5%) acorn barnacles.

Niche Type	No, inspected/ No. present	Biofouling Level (C/P/S/T) ⁵	Biofouling in excess of CRMS detected? (Y/N)	Comments
Dry-Dock Support Strips (DDSS)	various	S	N	DDSS mostly clean with only diatomaceous slime and occasional examples of either tubeworms or acorn barnacles (<1%).
Ballast plugs	various	S	N	DDSS mostly clean with occasional examples of either tubeworms or acorn barnacles (<1%).

5.4 Confidence Level

A number of environmental and logistical factors influence an inspectors' confidence of being able to successfully detect IMS of concern (if they are present amongst biofouling) during an inspection. X has developed criteria for assessing this confidence level, see Appendix 5. Confidence level does not refer to the statistical probability of IMS of concern being present on the vessel prior to, during, or after the inspection. It ensures our inspectors maintain a consistent inspection standard, and facilitates a transparent appraisal process for clients.

Criteria		Score	Maximum Possible
1) Vessel voyage history		5	5
2) Submerged hull inspection		6	14
3) Diving conditions		5	9
4) Quality of inspection		21	26
TOTAL		37	54
Confidence Level Achieved			
Low (0-20)	Moderate (21-30)	High (31-40)	Very High (41-54)
At the completion of the biofouling inspection, the Biofouling Inspector achieved a "high" level of confidence that macroscopic IMS of concern or viable biofouling in excess of the CRMS thresholds would have been detected had any been present during the inspection.			

5.5 Limitations and Declarations

The following limitations of the inspection are acknowledged. Quality assurance declarations are provided in the Appendices as appropriate.

Limitation or Declaration	Acknowledged
The inspection was unable to detect microscopic or juvenile life stages of IMS of concern.	☒ Yes ☐ No ☐ N/A
It is possible that IMS of concern could have settled on the vessel during or immediately after the inspection or in-water clean.	☒ Yes ☐ No ☐ N/A
If excessive biofouling was present, the inspection may not have been able to detect certain IMS, particularly if they were in very low abundance.	☐ Yes ☐ No ☒ N/A
The inspection only focussed on target IMS of concern and biofouling thresholds on the submerged external hull. Hence, biofouling or IMS of concern in ballast tanks was not included in this Scope of Works. It is therefore up to the vessel to independently comply with relevant ballast water management requirements.	☒ Yes ☐ No ☐ N/A
The client requested that suspected IMS detected during the inspection be confirmed by taxonomic experts or genetic confirmation.	☐ Yes ☐ No ☒ N/A
This report considers information received from third parties including the client. X proceed on the basis that the information supplied is accurate and complete, and has not sought to independently verify such information. X assumes no responsibility for any reporting conclusions or subsequent actions influenced by the provision of inaccurate or incomplete information to X.	☒ Yes ☐ No ☐ N/A
The X Diving Quality Assurance Declaration has been signed by the commercial diving company responsible for conducting the in-water inspection and/or clean (see Appendix 6).	☒ Yes ☐ No ☐ N/A

6. DISCUSSION

During the in-water biofouling inspection of the *MV Example*, the hull was found to be in good condition with biofouling on the general hull typically consisting of primary biofouling composed of diatomaceous slime and a waterline band of green algae. Secondary biofouling on the hull plating was largely restricted to the forward sections of the waterline (forward of Frame 198), the aft sections of waterline (aft of Frame 14) and the starboard quarter of the transom. This secondary biofouling was associated with a band of green algae extending to approximately 5 m depth and consisted of scattered barnacles (c.f. *Amphibalanus reticulatus* <5%) and tubeworms (*Hydroides* sp. <1%), with biofouling concentrated on weld seams. Some scattered secondary biofouling consisting of scattered individual acorn barnacles and tubeworms (<1%) was detected over other areas of the hull. A thorough inspection of niche areas determined that biofouling in niche areas was typically restricted to secondary biofouling composed of acorn barnacles (c.f. *Amphibalanus reticulatus*), tubeworms (*Hydroides* sp.), hydroids and bryozoans. In most cases biofouling in niche areas represented less than 5% coverage. However, biofouling associated with overboard pipes, draft markers (forward and aft), thruster propeller blades, anodes associated with the port stabiliser fin, rope guard voids and rudder stock areas exceeded this level. Secondary biofouling in some niche areas represented up to 90% coverage. Tertiary biofouling composed of scattered individuals of mussels (*Mytilus* sp.) was restricted to the rudder stock areas and a single individual associated with a port rope cutter blade (individual removed). The X Inspector achieved a high level of confidence that IMS of concern would have been detected if present and that biofouling in excess of the CRMS thresholds were accurately described by the inspection.

No macroscopic IMS of concern listed in regulatory requirements pertaining to Australian state legislation were detected in association with the *MV Example*. Thus, at the conclusion of the inspection, the vessel was determined to be compliant with the regulatory obligations of the Northern Territory, Queensland, New South Wales, Victoria and Tasmania. However, it should be recognised that microscopic life-history stages may have been present and not detected, and that individuals of IMS of concern may settle on the vessel's submerged hull subsequent to the inspection. To minimise the chance of infection the *MV Example* should limit exposure to IMS propagules by minimizing time spent in coastal waters.

The New Zealand Craft Risk Management Standard (CRMS) currently applies to all vessels entering New Zealand territory from the point when they enter the seaward boundary of the territorial sea (12 nautical miles from the coastal baseline), and for all the time they remain in the territory including in internal waters. In essence, the CRMS requires that all vessels visiting New Zealand territorial waters have a "clean hull", with the acceptable thresholds for this status being defined according to the intended length of stay in New Zealand. Two categories of vessels are distinguished in the CRMS:

- 1) Long-stay vessels' means those vessels intending to remain in New Zealand for 21 days or longer, or those vessels intending to visit areas other than those designated under Section 37 of the Biosecurity Act 1993 as 'Places of First Arrival'.
- 2) 'Short-stay vessels' means those vessels intending to remain in New Zealand for 20 days or less and to only visit places designated under section 37 of the Act as 'Places of First Arrival'. These vessels generally remain under 'biosecurity surveillance' while in New Zealand territory rather than becoming fully cleared of risk goods.

As the *MV Example* will be visiting various locations in New Zealand not designated under Section 37 the vessel would typically be required to be compliant with the "long-stay" threshold. However, if a Craft Risk Management Plan (CRMP) is prepared and approved for the vessel, the Ministry for Primary Industries (MPI) have indicated that compliance with the "short-stay thresholds" may be acceptable for cruise operators. To facilitate entry to New Zealand, X strongly recommends that Wilhelmsen consider developing a CRMP for biofouling.

An assessment of the *MV Example* determined that the vessel is not presently compliant with the CRMS "long-stay" threshold. Furthermore, various hull and niche locations have biofouling levels that are considered non-compliant with the CRMS "short-stay" thresholds. If Company X intends to demonstrate that the *MV Example* is compliant with the CRMS "short-stay" thresholds, some level of in-water cleaning is necessary, followed by an in-water biofouling inspection to determine that the vessel is compliant with appropriate biofouling thresholds prior to entering New Zealand territorial waters. **Table 3** provides a summary of areas on the hull of the *MV Example* exceeding the CRMS "short-stay" threshold, and **Figure 2** provides the approximate locations of all areas requiring cleaning. Cleaning advice is included in **Table 3**.

In facilitating any in-water cleaning operations Company X should consider two potential approaches to removing the problematic biofouling identified during this inspection:

- 1) *Comprehensive clean*

This approach requires the removal of tunnel thruster gratings to facilitate diver access and clean using high pressure wand or cavi-blaster. This process will require more time than typically available during an eight hour port turn-around.

- 2) *Opportunistic clean*

This approach would facilitate the cleaning of tunnel thruster propeller blades using long high-pressure wands operated by divers through the thruster gratings. Though consultation with a commercial dive company will be necessary, it is possible this could be conducted within the timeframe of an eight hour port turn-around.

Table 3. Submerged hull areas of the *MV Example* requiring remedial cleaning and suggested cleaning approach.

#	Hull Location	Description	Cleaning approach
1	Waterline	Forward section of waterline (forward of Frame 198), and aft section of waterline (aft of Frame 14) with <5% scattered barnacles and tubeworms in a band extending to approximately 5 m. Secondary biofouling concentrated on weld seams. Transom waterline with <5% scattered barnacles and tubeworms restricted to starboard quarter.	Cleaning using a diver operated brush kart will be the most time-effective approach.
2	Forward and Aft Draft markers	Biofouling consisting of <5% acorn barnacles and tubeworms associate with forward draft markers extending to depth of 5 m. Draft markers deeper than 5 m with isolated acorn barnacles (<1%).	If using a brush kart for the waterline this area should be cleaned simultaneous to the waterline area. Otherwise can be cleaned using high pressure wand or cavi-blaster.
3	Overboard pipes	All inspected overboard pipes found to have significant biofouling consisting of 60 – 90% acorn barnacles, tubeworms, bryozoans and hydroids.	Clean using high pressure wand or cavi-blaster.
4	Blades of all tunnel thrusters	Thruster propellers with ~50% biofouling consisting of acorn barnacles, tubeworms, hydroids and algae.	If possible remove hinged grating to facilitate diver access to clean using high pressure wand or cavi-blaster. If access is not possible, divers should attempt to clean all accessible surfaces using long high pressure wands through the grating.
5	Stabilising fin anode	One anode associated with port side stabiliser fin void with ~50% biofouling consisting of tubeworms, hydroids and acorn barnacles.	Clean using high pressure wand or cavi-blaster.
6	Rope guards and rope cutter voids (propeller and shaft)	A single juvenile mussel (<i>Mytilus</i> sp.) was detected and removed from a port side rope cutter blade associated with the shaft. Void spaces of shaft and propeller rope guards with 40-60% secondary biofouling consisting primarily of tubeworms with scattered (~5%) acorn barnacles.	Clean using high pressure wand or cavi-blaster. Cleaning may be restricted to accessible areas however large inspection ports should facilitate thorough clean.
7	Interface surfaces of propellers	Trailing face of propeller boss with 70-90% secondary biofouling composed primarily of tubeworms with scattered acorn barnacles. Rudder facings with ~10% tubeworms with scattered acorn barnacles.	Clean using high pressure wand or cavi-blaster. Rudders should be positioned to open interface gap to facilitate cleaning.
8	Rudders	Rudder hinges and void spaces with ~60% secondary biofouling consisting of acorn barnacles and tubeworms. Anodes with 20-40% secondary biofouling consisting of acorn barnacles and tubeworms.	Clean using high pressure wand or cavi-blaster.

		Non-toxic surfaces associated with top of rudder and rudder stock with 20-60% secondary biofouling. Scattered individual mussels (<i>Mytilus</i> sp.) present around rudder stock of both rudders (~10%).	
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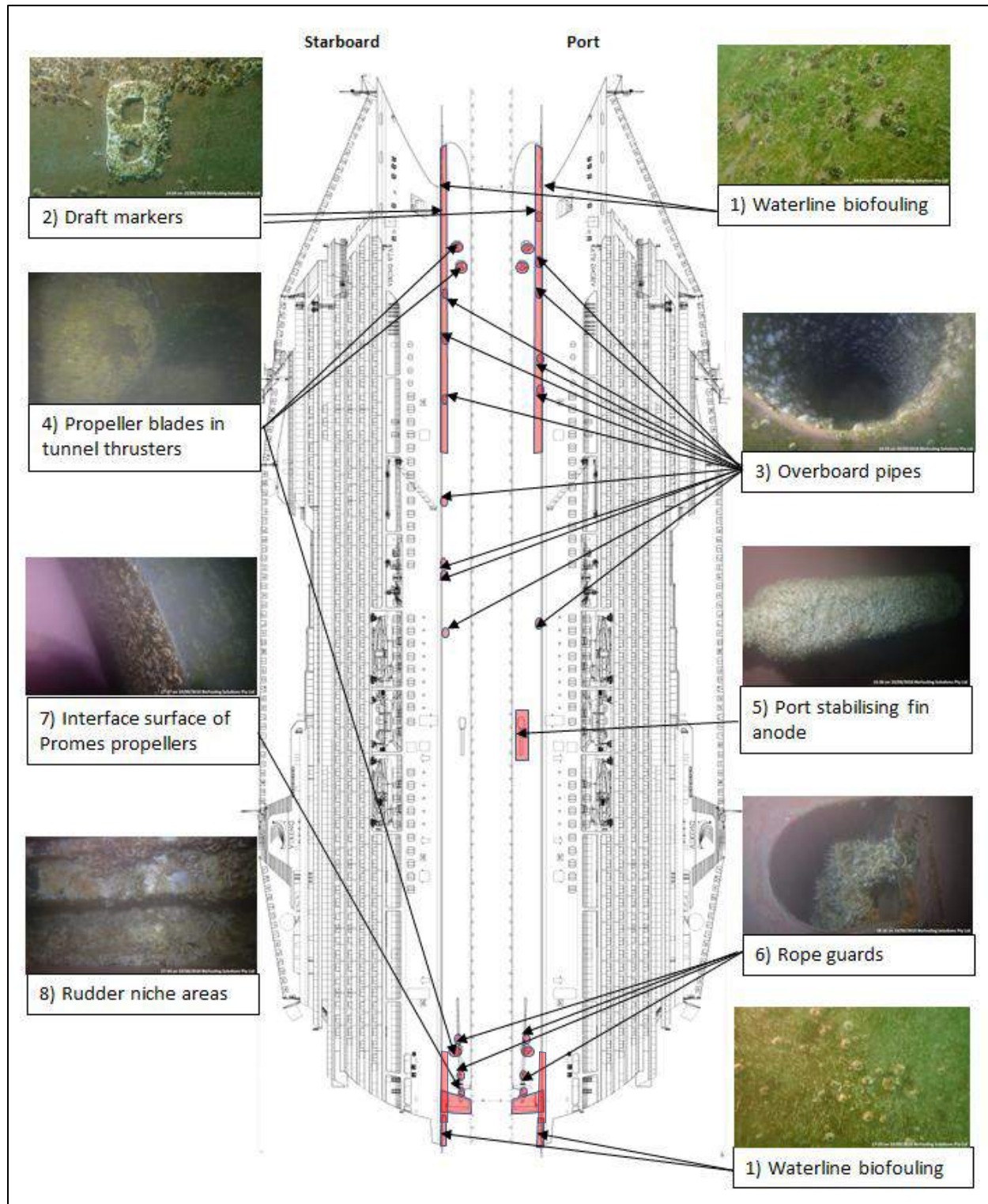


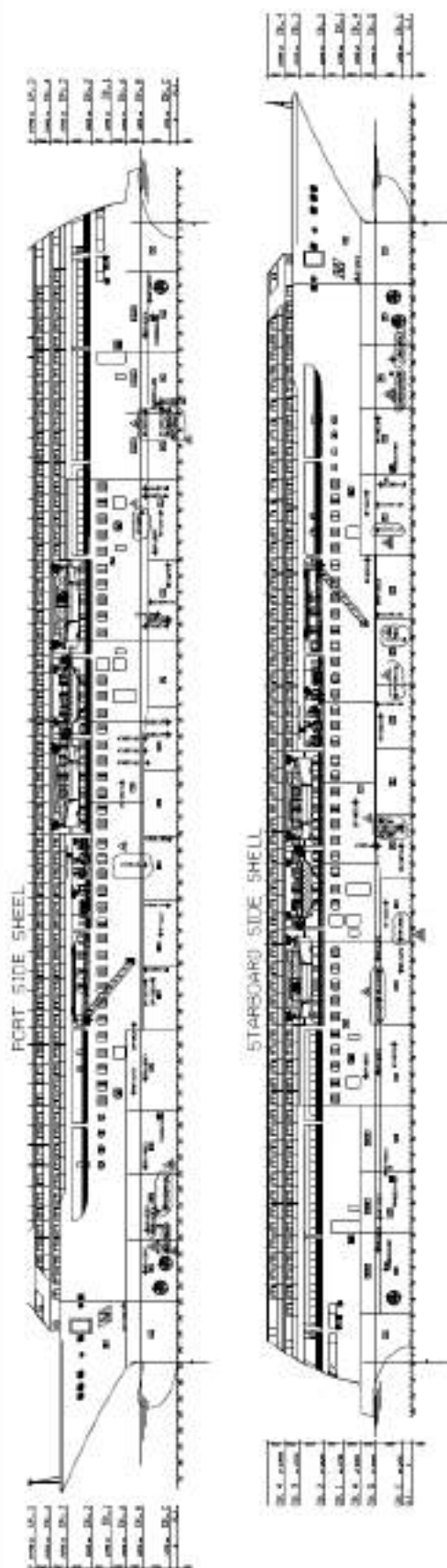
Figure 2 Submerged areas of the *MV Example* requiring remedial cleaning and examples of biofouling in excess of CRMS thresholds.

Company X should engage directly with suitable commercial dive companies to assess capabilities and availability to complete this scope of work within the available time frames in destination ports prior to the *MV Example* visiting New Zealand. X has been advised that Company X is considering undertaking the clean in Hong Kong (3-5 November, 2018), Laem Chabang (17-19 November, 2018) or Singapore (23 November, 2018). While it may be possible to facilitate an opportunistic clean in Singapore on 23 November 2018, the risks associated with the short turn around and the possibility of simultaneous operations hampering the dive effort (e.g. bunkering, passenger operations etc.) may result in a high risk of the scope of work not being completed. Unfortunately, X has limited experience with commercial dive operators in Laem Chabang or Hong Kong but it is likely that divers based in these locations will not be trained in biofouling management to a similar level to those in Singapore. An alternative may be to complete the cleaning in either Laem Chabang or Hong Kong and then engage a commercial dive company in Singapore to conduct a final inspection on 23 November with divers standing by to clean any residual biofouling not removed during the previous dive operation.

7. REFERENCES

- Carlton, J. T., 1985. Transoceanic and interoceanic dispersal of coastal marine organisms: the biology of ballast water. *Oceanography and Marine Biology Annual Review*, 23: 313-371.
- Carlton, J. T., 1996. Pattern, process, and prediction in marine invasion ecology. *Biological Conservation*, 78: 97-106.
- Carlton, J. T., 2001. Introduced species in U. S. coastal waters: environmental impacts and management priorities, Arlington, Virginia, United States, Pew Oceans Commission, 28 p.
- Eldredge, L. G., Carlton, J. T., 2002. Hawaiian Marine Bioinvasions: A Preliminary Assessment. *Pacific Science*, 56:211-212.
- Hewitt, C. L., 2003. The diversity of likely impacts of introduced marine species in Australian waters. *Records of the South Australian Museum Monograph Series* 7, 3-10.
- Hewitt, C.; Campbell, M., 2010. The relative contribution of vectors to the introduction and translocation of invasive marine species, Canberra City, The Department of Agriculture, Fisheries and Forestry.
- Hewitt, C.; Campbell, M.; Thresher, R; Martin, R., 1999. Marine Biological Invasions in Port Phillip Bay, Victoria. CSIRO Centre For Research on Introduced Marine Pests (Ed.) Technical Report No. 20.
- Hewitt, C.; Campbell, M.; Thresher, R.; Martin, R.; Boyd, S.; Cohen, B.; Currie, D.; Gomon, M.; Keogh, M.; Lewis, J.; Lockett, M.; Mays, N.; Macarthur, M.; O'Hara, T.; Poore, G.; Ross, D.; Storey, M.; Watson, J.; Wilson, R., 2004. Introduced and cryptogenic species in Port Phillip Bay, Victoria. *Marine Biology*, 144: 183-202.
- Hewitt, C.; Campbell, M.; Coutts, A.; Dahlstrom, A.; Shields, D.; Valentine, J., 2011. Species Biofouling Risk Assessment, Canberra, Australia, Department of Agriculture, Fisheries and Forestry.
- Pimentel, D.; Lach, L.; Zuniga, R.; Morrison, D., 2000. Environmental and economic costs of nonindigenous species in the United States. *BioScience* 50: 53-65.
- Thresher, R. E.; Hewitt, C. L.; Campbell, M. L., 1999. Synthesis: Introduced and cryptogenic species in Port Phillip Bay. In: Hewitt, C. L.; Campbell, M. L.; Thresher, R. E; Martin, R. B., (Eds.), Marine biological invasions of Port Phillip Bay, Victoria. Centre for Research on Introduced Marine Pests. Technical Report No. 20: 283-295.

Appendix 1 General Arrangement



Appendix 2 Antifouling Certificate



CERTIFIED COPY

Certificate no: **VEN 1890876**
Page 1 of 2

International Anti-Fouling System Certificate

(This certificate shall be supplemented by a Record of Anti-Fouling Systems)
Issued under the International Convention on the Control of Harmful Anti-Fouling Systems on Ships
under the authority of the Government of the Kingdom of Norway by Lloyd's Register EMEA.

When a Certificate has been previously issued this Certificate replaces the certificate dated **Not Applicable**

	Particulars of ship
Name of ship	
Port of registry	
Distinctive numbers or letters	
Gross tonnage	
IMO number	

An anti-fouling system controlled under Annex 1 has not been applied during or after construction of this ship. ☒

An anti-fouling system controlled under Annex 1 has been applied on this ship previously, but has been removed by _____ or _____. ☐

An anti-fouling system controlled under Annex 1 has been applied on this ship previously, but has been covered with a sealer coat applied by _____ on _____. ☐

An anti-fouling system controlled under Annex 1 was applied on this ship prior to _____ but will be removed or covered with a sealer coat prior to _____. ☐

This is to certify that

1. the ship has been surveyed in accordance with regulation 1 of Annex 4 to the Convention; and
2. the survey shows that the anti-fouling system on the ship complies with the applicable requirements of Annex 1 to the Convention

Issued at **Venice** on **07 June 2018**
Date of completion of the survey on which this certificate is issued **27 September 2017**



A. Solano
Surveyor to Lloyd's Register EMEA
a member of the Lloyd's Register group

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Form 1425 (2/01/05)

CERTIFIED COPY

Certificate no:
Page 2 of 2

VEN 1690876

Record of Anti-Fouling Systems

This record shall be permanently attached to the International Anti-Fouling System Certificate

	Particulars of ship
	Name of ship
	Distinctive numbers or letters
	IMO number
Type(s) of anti-fouling system(s) used	Details of anti-fouling system applied Biocidal
Date(s) of application of anti-fouling system(s)	25 September 2017
Name(s) of company(ies) and facility(ies)/location(s) where applied	Fincantieri S.p.A. - Ancona Shipyard
Name(s) of anti-fouling system manufacturer(s)	Jotun Paints (Europe) Ltd., Stather Road, Flakborough, Scunthorpe, UK,
Name(s) and colour(s) of anti-fouling system(s)	SeaQuantum Ultra 5 - Dark Red
Active ingredient(s) and their Chemical Abstract Service Registry Number(s) (CAS Number(s))	CD - Cuprous Oxide - 1317-39-1 CP - Cuprous Pyrrithione - 14815-37-8
Type(s) of sealer coat, if applicable	Not Applicable
Name(s) and colour(s) of sealer coat applied, if applicable	Not Applicable
Date of application of sealer coat	Not Applicable

This is to certify that this record is correct in all respects

Issued at: **Venice**

on: **07 June 2018**

Lloyd's Register EMEA
A member of the Lloyd's Register Group



A. Solano
Venice Office
A. Solano

Surveyor to Lloyd's Register EMEA

a member of the Lloyd's Register group.

Form 1425 (2015-08)

Appendix 3 Levels of Biofouling

The Commonwealth of Australia has devised a simplistic, but pragmatic method of assessing the likelihood of vessels or infrastructure being infested by IMS of concern. It relates likelihood of an IMS being present to the level of complexity in the biofouling assemblage. Levels of biofouling tend to increase the longer a vessel or structure remains submerged in seawater. Three levels of biofouling are identified on the basis of the organisms present. The likelihood of IMS of concern being present amongst biofouling increases markedly as the assemblage develops from primary to secondary and finally to tertiary levels.

The colonisation and accumulation of biofouling on surfaces submerged in seawater depend on a series of complex processes. Primary biofouling of submerged surfaces begins with biochemical and bacterial conditioning the moment a vessel or structure is immersed in seawater. This is followed by colonisation by diatoms, protozoans and multi-cellular organisms. This conditioning and colonisation of surfaces by microscopic organisms provides an ideal substrate for settlement and growth of more visible organisms such as fine filamentous algae, (some of which are resistant to the toxic biocides contained in anti-fouling coatings). In turn their establishment provides a suitable, (but not necessarily critical) substrate for settlement of the secondary biofouling organisms which are most commonly encountered during inspections. Secondary biofouling is more likely to develop into tertiary biofouling in niche areas of vessels or structures that are protected from strong hydrodynamic forces, when vessels remain stationary for long periods of time, or as the in-service period for the vessel increases.

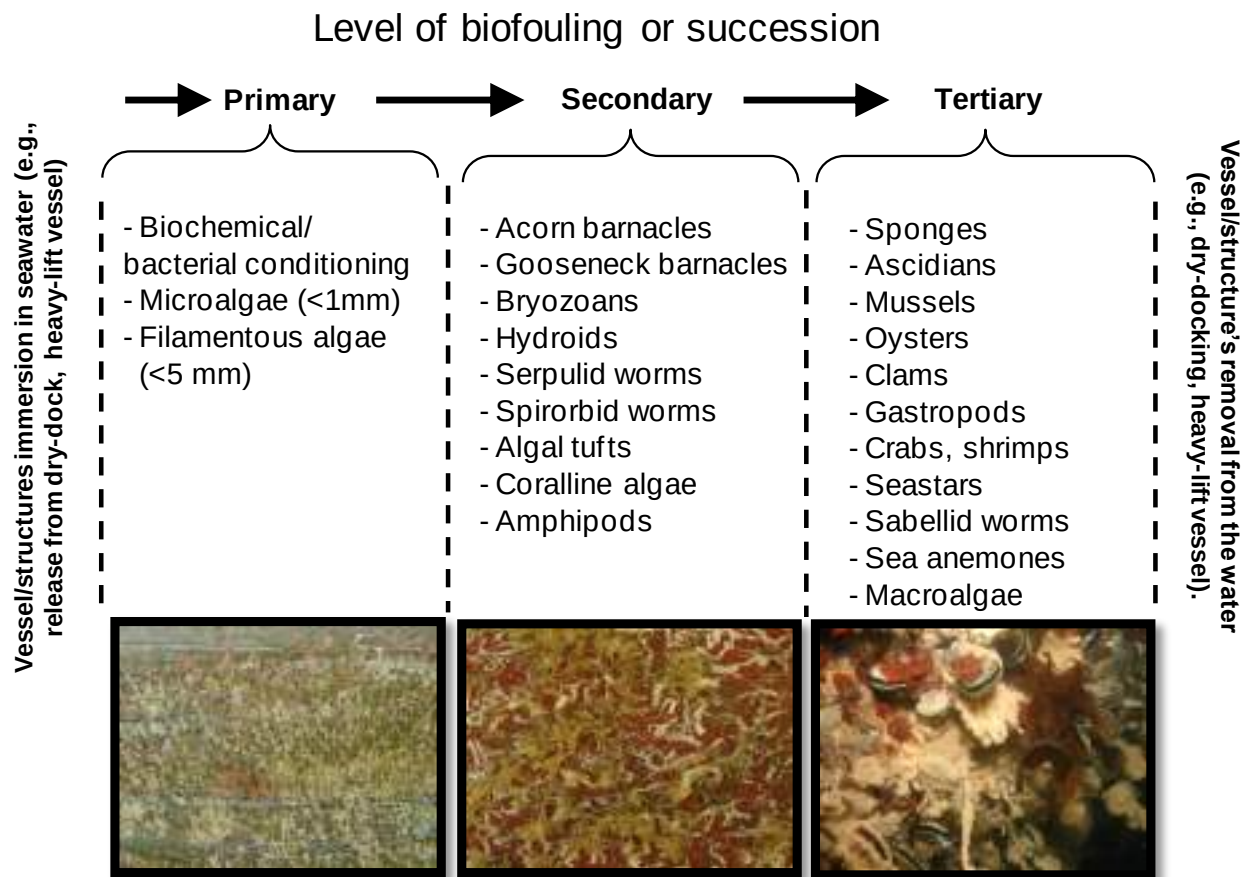


Figure 3. The simplistic, but pragmatic approach to assessing IMS likelihood using three levels of biofouling.

Appendix 4 Final External Hull Inspection Photos

Photos in this appendix are representative of the locations and equipment inspected. Further photos can be made available on DVD upon request.



Plate 1 Acorn barnacles >1% associated with weld seam. Forward of Frame 198, port, waterline to 5m.



Plate 2 Band of acorn barnacles >1%. Forward of Frame 198, port, waterline to 5m.

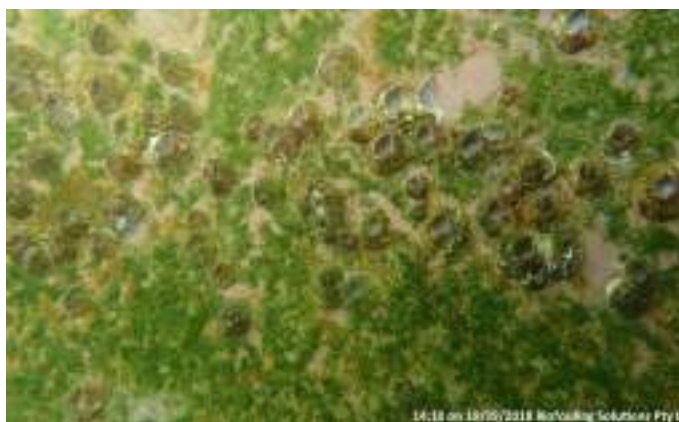


Plate 3 Band of acorn barnacles >1% associated with waterline. Forward of Frame 198, port, waterline to 5m.



Plate 4 Band of green algae. Port, waterline.



Plate 5 Interface between band of biofouling and clean hull. Forward of Frame 198, port, Approximately 5m depth.



Plate 6 Clean hull below 5m. Forward of Frame 198, port.



Plate 7 Band of acorn barnacles >1%. Forward of Frame 198, starboard, waterline to 5m.



Plate 8 Acorn barnacles and tubeworms >1% associated with weld seam. Forward of Frame 198, starboard, waterline to 5m.



Plate 9 Clean hull below 5m. Forward of Frame 198, starboard.



Plate 10 Band of green algae without secondary biofouling. Starboard waterline aft of Frame 198.



Plate 11 Port side forward draft markers showing secondary biofouling > 5%.



Plate 12 Port side forward draft markers deeper than 5m with no secondary biofouling.



Plate 13 Starboard side forward draft markers showing secondary biofouling > 5%.



Plate 14 Starboard side forward draft markers deeper than 5m with no secondary biofouling.



Plate 15 Depth sounder with <5% secondary biofouling restricted to bolts. Forward amidships.



Plate 16 Detail of secondary biofouling restricted to bolts of depth sounder. Forward amidships.



Plate 17 Speed log with <5% secondary biofouling. Forward amidships.



Plate 18 Flat bottom shell plating. Forward amidships.



Plate 19 Flat bottom shell plating. Amidships.



Plate 20 Dry Dock Support Strip, amidships.



Plate 21 Dry Dock Support Strip, amidships.



Plate 22 Dry Dock Support Strip, amidships.



Plate 23 Amidships waterline algae free of secondary biofouling. Port.



Plate 24 Amidships waterline algae free of secondary biofouling. Starboard.



Plate 25 Port side amidships draft markers showing secondary biofouling $\leq 5\%$.



Plate 26 Port side amidships draft markers showing secondary biofouling $< 5\%$.



Plate 27 Port aft shell plating with secondary biofouling $>1\%$ associated with weld seam (aft of Frame 14).



Plate 28 Port aft shell plating with secondary biofouling $>5\%$ associated with weld seam (aft of Frame 14).



Plate 29 Port side aft draft markers showing secondary biofouling $> 5\%$.



Plate 30 Port side aft draft markers showing secondary biofouling $> 5\%$.



Plate 31 Transom showing secondary biofouling >1% (restricted to starboard quarter). NB: boot-top below waterline.



Plate Transom showing secondary biofouling >1% (restricted to starboard quarter). NB: boot-top below waterline.



Plate 33 Starboard aft shell plating with secondary biofouling >1% associated with weld seam (aft of Frame 14).



Plate 34 Starboard aft shell plating with secondary biofouling >1% associated with weld seam (aft of Frame 14).



Plate 35 Starboard side aft draft markers showing secondary biofouling $\geq$ 5%.



Plate 36 Starboard side aft draft markers showing secondary biofouling $\geq$ 5%.



Plate 37 Selection of overboard pipes showing secondary biofouling >5%.



Plate 38 Selection of overboard pipes showing secondary biofouling >5%.



Plate 39 Selection of overboard pipes showing secondary biofouling >5%.



Plate 40 Selection of overboard pipes showing secondary biofouling >5%.



Plate 41 Selection of overboard pipes showing secondary biofouling >5%.



Plate 42 Selection of overboard pipes showing secondary biofouling >5%.



Plate 43 Area of damages paint, turn of bilge, starboard, forward.

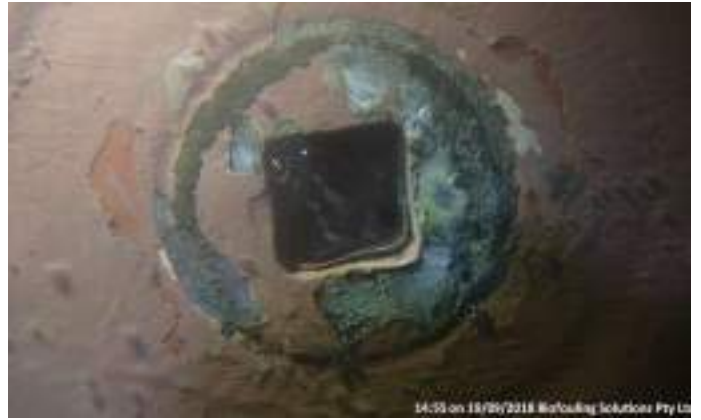


Plate 4 Ballast plug, flat bottom amidships.



Plate 45 Ballast plug, flat bottom amidships.



Plate 46 Ballast plug, flat bottom amidships.



Plate 47 Starboard side bilge keel, leading edge.



Plate 48 Starboard side bilge keel, upper surface.



Plate 49 Starboard side bilge keel, lower surface.



Plate 50 Port side bilge keel, leading edge.



Plate 51 Port side bilge keel, upper surface.



Plate 52 Bow thruster #1 grating. Starboard side.



Plate 53 Bow thruster #1 grating. Starboard side.



Plate 54 Bow thruster #1 grating. Port side showing secondary biofouling <5%.



Plate 55 Bow thruster #1. Anode and prop viewed through grating. Divers observed ~50% Secondary biofouling on prop and boss.



Plate 56 Bow thruster #1. Gear housing viewed through grating.



Plate 57 Bow thruster #1. Mounting support viewed through grating.



Plate 58 Bow thruster #1. Anode and tunnel wall viewed through grating.



Plate 59 Bow thruster #2 grating. Port side.



Plate 60 Bow thruster #2 grating. Port side.



Plate 61 Bow thruster #2 grating. Starboard side.



Plate 62 Bow thruster #2. Anode and prop viewed through grating.

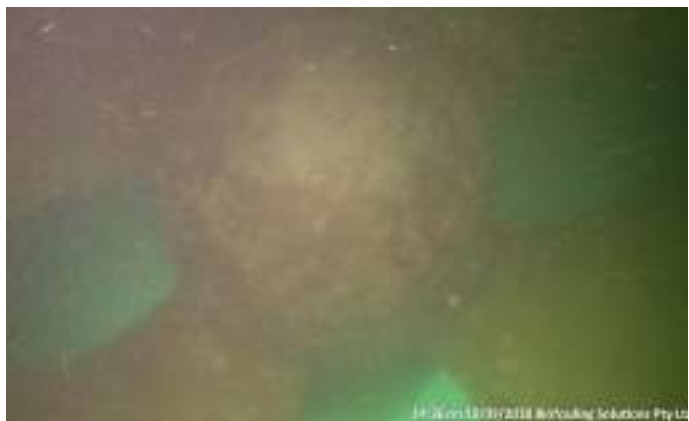


Plate 63 Bow thruster #2. Propeller viewed through grating. Divers observed ~50% Secondary biofouling on prop and boss.

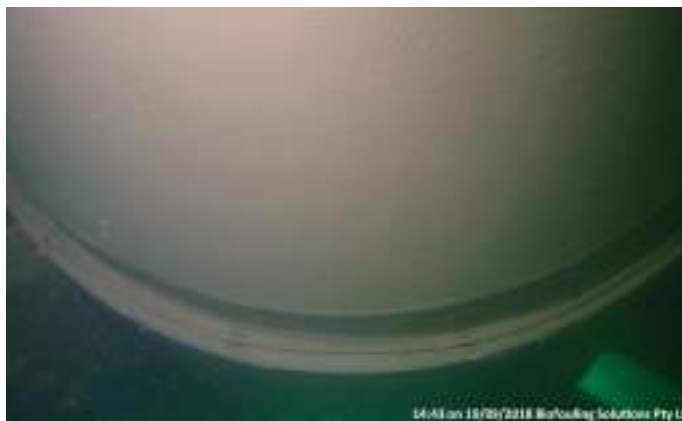


Plate 64 Bow thruster #2. Gear housing viewed through grating.



Plate 65 Bow thruster #2. Mounting support viewed through grating.



Plate 66 Bow thruster #2. Anode and tunnel wall viewed through grating.



Plate 67 Port side stabilising fin. Inspected in retracted position.



Plate 68 Anode with secondary biofouling (>5%) on port side stabilising fin. Inspected in retracted position.



Plate 69 Port side stabilising fin. Inspected in retracted position.



Plate 70 Port side stabilising fin. Inspected in retracted position.



Plate 71 Port side stabilising fin. Inspected in retracted position.



Plate 72 Clean anode on port side stabilising fin. Inspected in retracted position.



Plate 73 Starboard side stabilising fin. Inspected in retracted position.



Plate 74 Starboard side stabilising fin. Inspected in retracted position.



Plate 75 Clean anode on Starboard side stabilising fin. Inspected in retracted position.



Plate 76 Starboard side stabilising fin. Inspected in retracted position.



Plate 77 Port rudder with >5% secondary biofouling. Top of rudder stock.



Plate 78 Port rudder with >5% secondary biofouling.



Plate 79 Port rudder anode with >5% secondary biofouling.



Plate 80 Port rudder anode with >5% secondary biofouling.



Plate 81 Top of port rudder stock with >5% secondary biofouling.



Plate 82 Top of port rudder stock with >5% secondary biofouling and scattered tertiary biofouling (mussels)



Plate 83 Port rudder hinge.



Plate 84 Port rudder plating.



Plate 85 Port Promes propeller showing interface surface with >5% secondary biofouling.



Plate 86 Port side propeller blades.



Plate 87 Port side rope guard void showing >5% secondary biofouling.

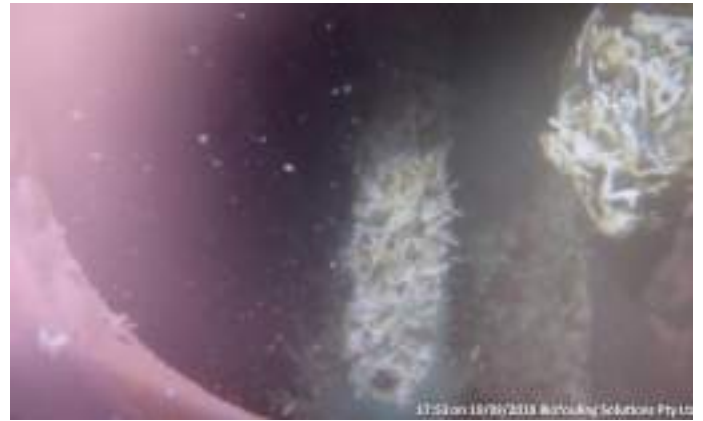


Plate 88 Port side rope guard void showing >5% secondary biofouling.



Plate 89 Port side rope guard void showing >5% secondary biofouling.



Plate 90 Port side rope guard void showing >5% secondary biofouling.



Plate 91 Forward high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard.



Plate 92 Forward high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard.



Plate 93 Forward high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard.



Plate 94 Forward high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard.



Plate 95 Amidships high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard.



Plate 96 Amidships high sea chest grating showing <5% secondary biofouling restricted to bolts. Starboard

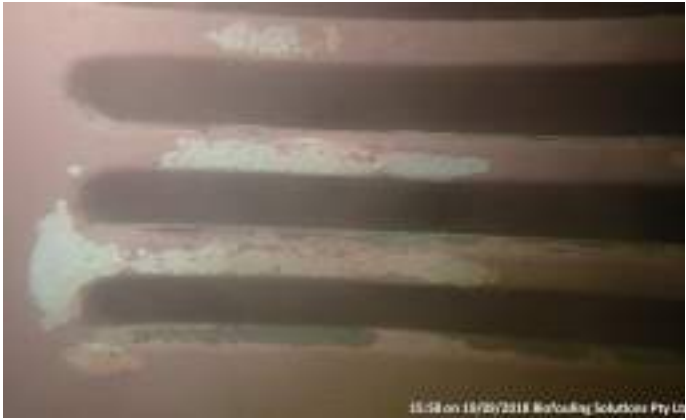


Plate 97 Aft low Sea chest grating. Port.

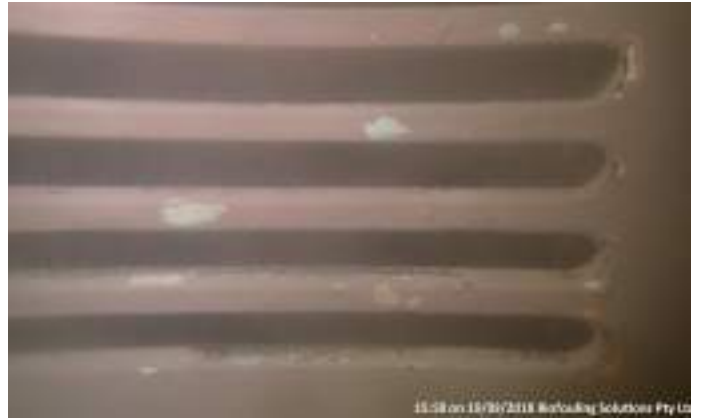


Plate 98 Aft low Sea chest grating. Port.

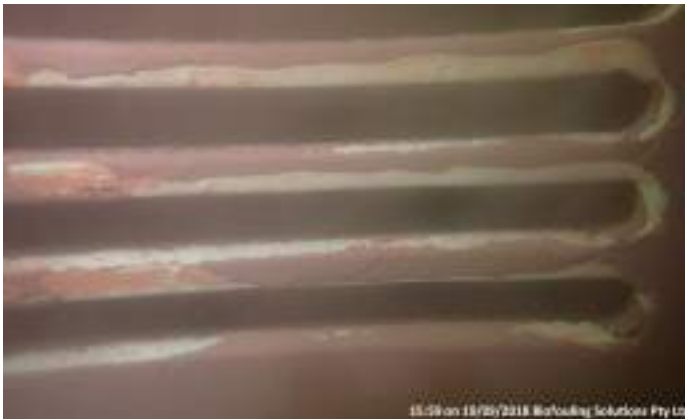


Plate 99 Aft low Sea chest grating. Port.



Plate 100 Aft low Sea chest grating. Port.



Plate 101 Aft low Sea chest grating. Port.



Plate 102 Aft low Sea chest grating with <1% secondary biofouling. Port.

Appendix 5 X Criteria to determine confidence level from vessel inspections

Appendix 6 X Diver Quality Assurance Declaration

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

Vessel/infrastructure's name:

Location: SINGAPORE CRUISE CENTRE Starting Date: 19-08-17

The Dive Supervisor declares that they understand and agree to the following separation of roles:

1. _____ supplies an Invasive Marine Species (IMS) inspector to:
 - A. Assess compliance of the vessel/infrastructure with the appropriate IMS management requirements. In order to assess compliance they may need to do the following:
 - B. Provide guidance on the standards and requirements of the inspection and/or the cleaning required for the vessel/infrastructure. This may involve requests, directions or instructions being provided to Divers via the Dive Supervisor.
 - C. Request the collection of biological samples and/or photographs from specified areas.
2. The Dive Supervisor is responsible for all aspects of diving safety, the coordination of divers and has the final decision on all aspects of the dive team's professional activities. The Dive Supervisor must:
 - A. Conduct appropriate risk assessments to ensure the safety of divers and diving operations.
 - B. Ensure procedures, equipment and dive team members are safe and appropriate for all aspects of the inspection and/or cleaning operations.
 - C. Be present at all times while Divers are undertaking the inspection or cleaning, and conduct communication with Divers underwater. In addition,
 - D. If the Dive Supervisor objects to the IMS Inspector's requests, directions or instructions they must promptly discuss all relevant issues with the IMS Inspector to safeguard the Divers.

The Dive Supervisor declares that they understand the IMS inspection and/or clean must achieve the standards stipulated by the following regulations and/or biofouling management requirements:

- ☐ Chevron Australia Pty Ltd's Barrow Island – Marine Vessels Procedure
- ☐ Woodside Energy Ltd's Invasive Marine Species Management Plan
- ☐ Western Australian Department of Fisheries *Fish Resources Management Act 1994*
- ☒ Other: NZ CRMS

Initials: Dive Supervisor: AS Contractor: _____ E

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

Treatment standard required:

- ☒ Level of Biofouling (i.e. Remove Primary, Secondary and/or Tertiary Levels of Biofouling)
☐ Remove Suspected Invasive Marine Species of Concern

Diving activity required:

- ☒ Inspection
☐ In-water clean

The under-signed declares that they and their divers took all reasonable steps to conduct the inspection and were willing and able to:

- ☒ Follow directions given by the Biofouling Solutions inspector
☒ Achieve the standards requested by the Biofouling Solutions inspector
☐ Conduct in-water cleaning treatments to ensure the complete removal of macroscopic secondary or tertiary biofouling and/or suspected Invasive Marine Species of Concern from the following locations:

N/A

The under-signed declares that diver's did not leave any areas where biofouling exceeded the aforementioned standard except for:

N/A

Initials: Dive Supervisor: AB Contractor: _____

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

These could not be cleaned because:

N/A

Additional Comments:

INSPECTION TO N2 CANS EXAMINING LEVEL OF
FILLING.

Initials: Dive Supervisor: *AS* Contractor:

DIVER SAFETY AND QUALITY ASSURANCE STATEMENT

The under-signed also declares that their divers conducted the final closed circuit television survey in such a manner that the _____ IMS inspector was able to accurately verify that the cleaning had achieved the aforementioned standard.

Commercial Diving Company: Singapore Salvage Engineers

Name: T.A. YONG Official capacity: Dive Supv.

Signed: [Signature] Date: 19/09/18

Contractor: _____

Name: _____ Official capacity: _____

Signed: _____ Date: _____

Name: _____ Official capacity: _____

Signed: _____ Date: _____